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Chapter 13	Wheeled Mobile Robots
	13.1 Types of Wheeled Mobile Robots
	13.2 Omnidirectional Wheeled Mobile Robots

Important concepts, symbols, and equations

Types of wheels



conventional



omniwheel



mecanum wheel

Kinematic wheeled
mobile robots
(no skidding, slipping)



differential drive



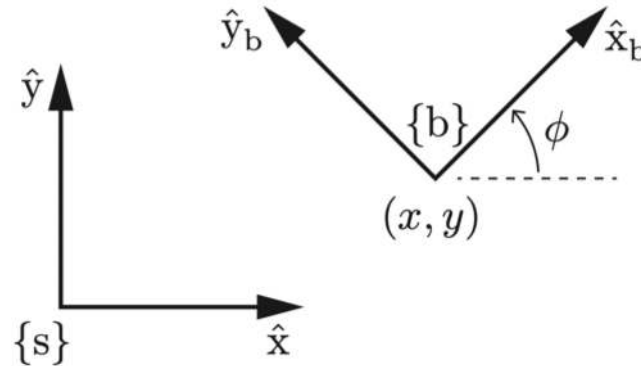
car-like

Ackermann steering



omnidirectional

Important concepts, symbols, and equations (cont.)



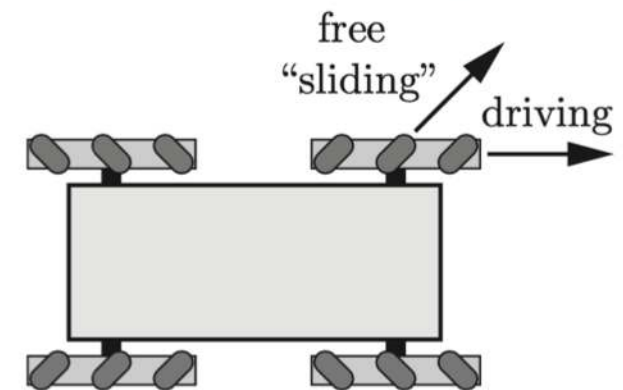
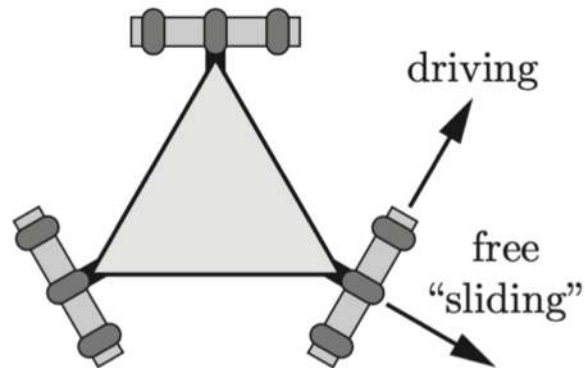
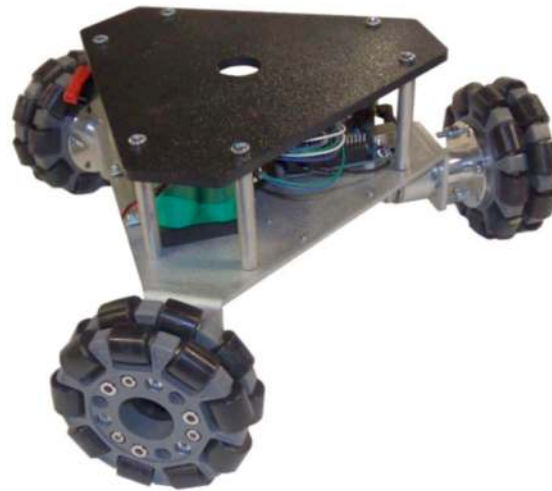
Configuration of the mobile base

$$T_{sb} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_1 \\ r_{21} & r_{22} & r_{23} & p_2 \\ \cancel{r_{31}} & \cancel{r_{32}} & \cancel{r_{33}} & \cancel{p_3} \\ 0 & 0 & 0 & 1 \end{bmatrix} \in SE(2) \text{ or } q = (\phi, x, y) \in \mathbb{R}^3$$

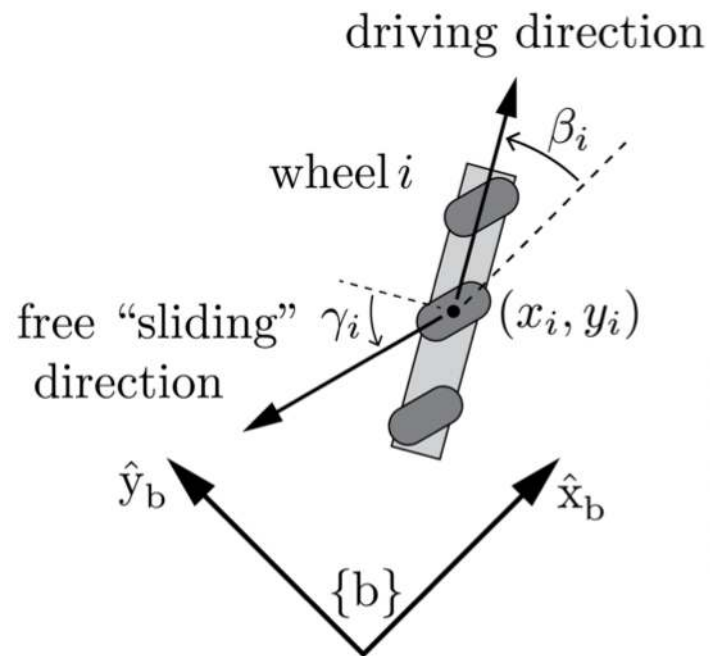
Velocity of the mobile base: $\mathcal{V}_b = (\cancel{\omega_{bx}}, \cancel{\omega_{by}}, \boxed{\omega_{bz}}, \cancel{v_{bx}}, \cancel{v_{by}}, \cancel{v_{bz}}) \text{ or } \dot{q} \in \mathbb{R}^3$

Important concepts, symbols, and equations (cont.)

Examples of
omnidirectional
wheeled mobile
robots



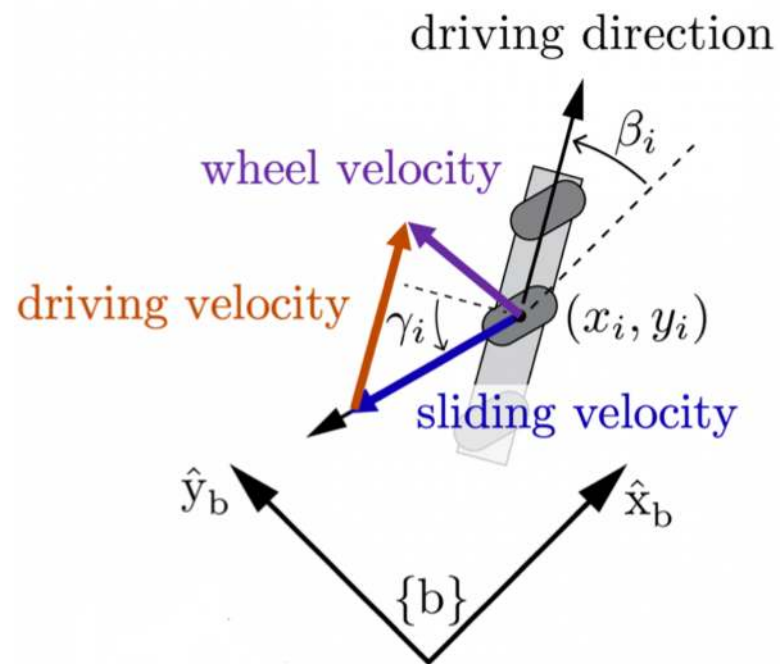
Important concepts, symbols, and equations (cont.)



wheel driving speed:

$$u_i = \frac{1}{r_i} [1 \quad \tan \gamma_i] \begin{bmatrix} \cos \beta_i & \sin \beta_i \\ -\sin \beta_i & \cos \beta_i \end{bmatrix} \begin{bmatrix} -y_i & 1 & 0 \\ x_i & 0 & 1 \end{bmatrix} \mathcal{V}_b$$

$\frac{1}{r_i}$: wheel radius
 $[1 \quad \tan \gamma_i]$: component in driving direction
 $\begin{bmatrix} \cos \beta_i & \sin \beta_i \\ -\sin \beta_i & \cos \beta_i \end{bmatrix}$: linear velocity at wheel, in wheel frame
 $\begin{bmatrix} -y_i & 1 & 0 \\ x_i & 0 & 1 \end{bmatrix}$: linear velocity at wheel, in $\{b\}$



$$u_i = h_i(0) \mathcal{V}_b$$

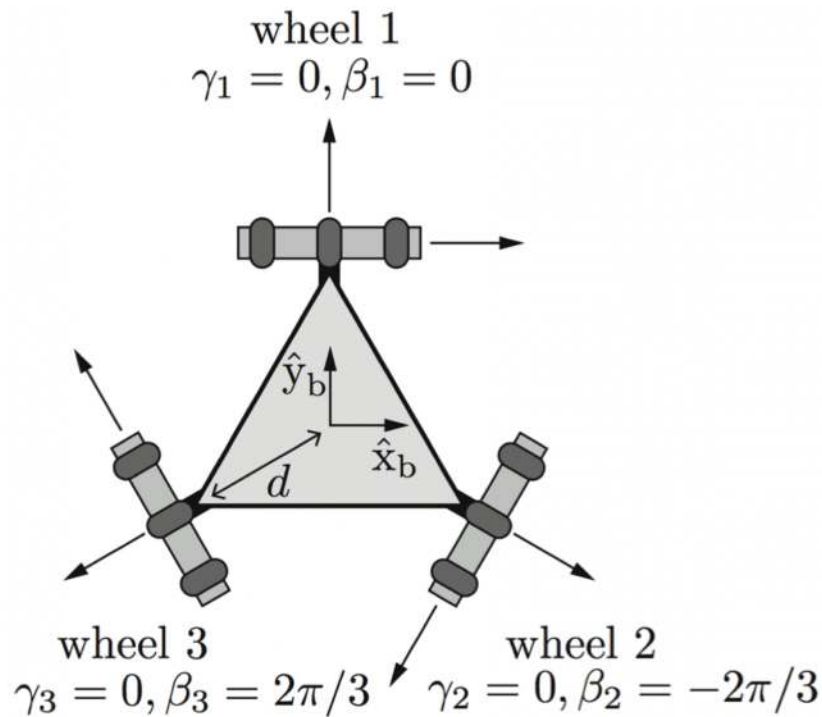
$$H(0) = \begin{bmatrix} h_1(0) \\ \vdots \\ h_m(0) \end{bmatrix} \in \mathbb{R}^{m \times 3}$$

m wheels

$$u = H(0) \mathcal{V}_b$$

wheel speed vector

Important concepts, symbols, and equations (cont.)



$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} = H(0)\mathcal{V}_b = \frac{1}{r} \begin{bmatrix} -d & 1 & 0 \\ -d & -1/2 & -\sin(\pi/3) \\ -d & -1/2 & \sin(\pi/3) \end{bmatrix} \begin{bmatrix} \omega_{bz} \\ v_{bx} \\ v_{by} \end{bmatrix}$$

spin in place

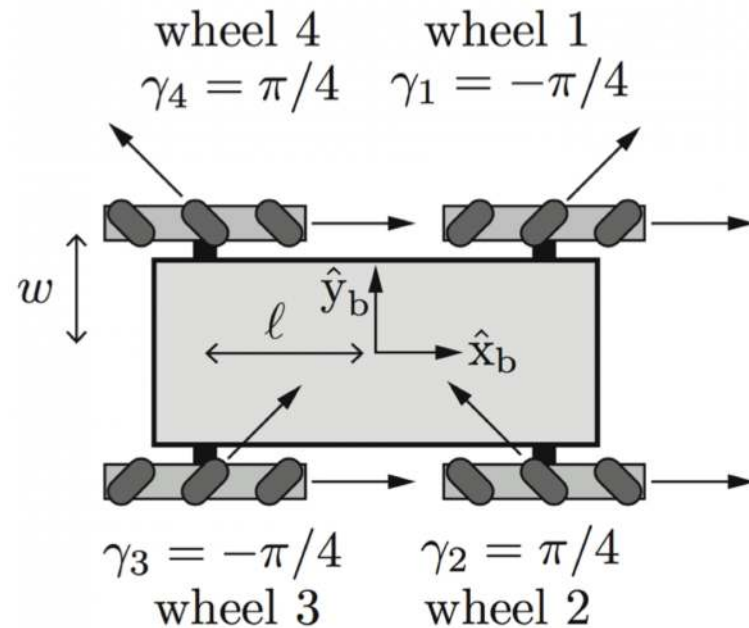
right

up

Why are at least three wheels required for omnidirectional motion?

Need at least 3 controls (wheel speeds) to map to all of the 3d twist space

Important concepts, symbols, and equations (cont.)



Wheel velocities are 4d.
Chassis twist is 3d.
Implications?

All u not in the 3d space $\{H(0)\mathcal{V}_b \text{ for all } \mathcal{V}_b\}$ mean that the wheels must be skidding

$$u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} = H(0)\mathcal{V}_b = \frac{1}{r} \begin{bmatrix} -\ell - w & 1 & -1 \\ \ell + w & 1 & 1 \\ \ell + w & 1 & -1 \\ -\ell - w & 1 & 1 \end{bmatrix} \begin{bmatrix} \omega_{bz} \\ v_{bx} \\ v_{by} \end{bmatrix}$$

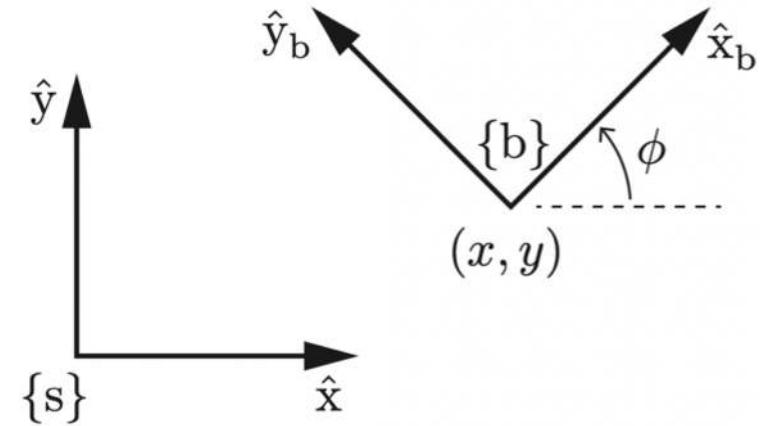
Important concepts, symbols, and equations (cont.)

Wheel speeds in terms of $\dot{q}$:

$$u = H(0)\mathcal{V}_b, \quad H(0) \in \mathbb{R}^{m \times 3}$$

$$u = H(0) \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \phi & \sin \phi \\ 0 & -\sin \phi & \cos \phi \end{bmatrix}}_{H(\phi)} \begin{bmatrix} \dot{\phi} \\ \dot{x} \\ \dot{y} \end{bmatrix} \quad \dot{q}$$

$$u = H(\phi)\dot{q}$$



Important concepts, symbols, and equations (cont.)

Feedforward + PI feedback stabilization of a planned trajectory:

$$\dot{q}(t) = \dot{q}_d(t) + K_p(q_d(t) - q(t)) + K_i \int_0^t (q_d(t) - q(t)) dt$$

$$u = H(\phi)\dot{q}$$

$K_p, K_i > 0$
for stability

Setpoint control:
 $e_{ss} = 0$ for P, PI

Constant velocity reference:
 $e_{ss} = 0$ for PI

Stability and steady-state error
for different control laws and
desired trajectories?

Important concepts, symbols, and equations (cont.)

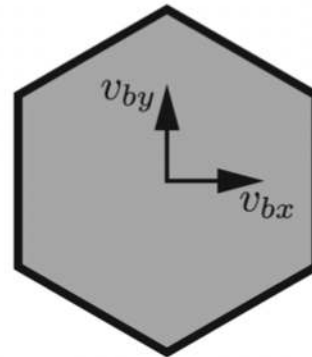
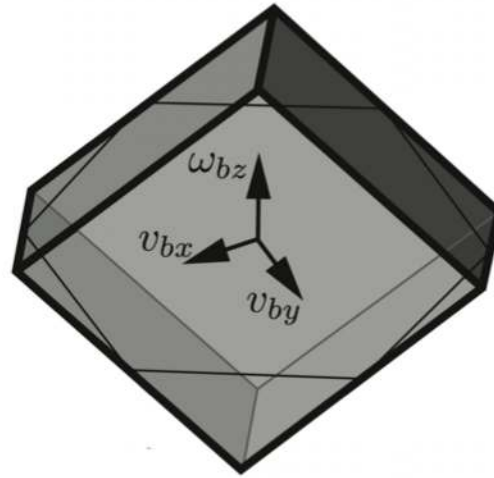
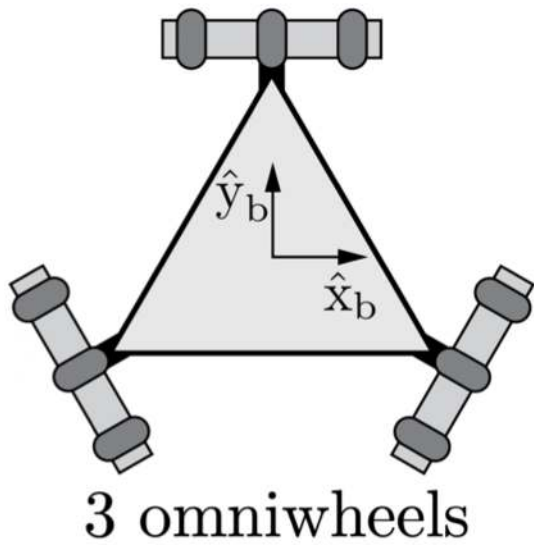
Given wheel velocity limits, the chassis' feasible twists lie inside a $2m$ -sided convex polyhedron:

$$\begin{bmatrix} u_1 \\ \vdots \\ u_m \end{bmatrix} = H(0)\mathcal{V}_b = \begin{bmatrix} h_1(0) \\ \vdots \\ h_m(0) \end{bmatrix} \mathcal{V}_b$$

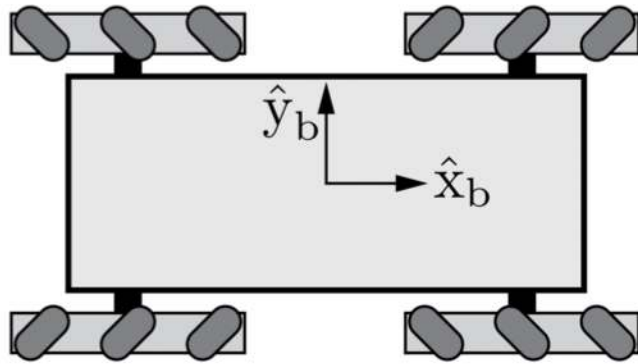
$$-u_{i,\max} \leq u_i = h_i(0)\mathcal{V}_b \leq u_{i,\max}$$

$$\left. \begin{array}{l} -u_{i,\max} = h_i(0)\mathcal{V}_b \\ u_{i,\max} = h_i(0)\mathcal{V}_b \end{array} \right\} \text{define two parallel bounding planes in twist space}$$

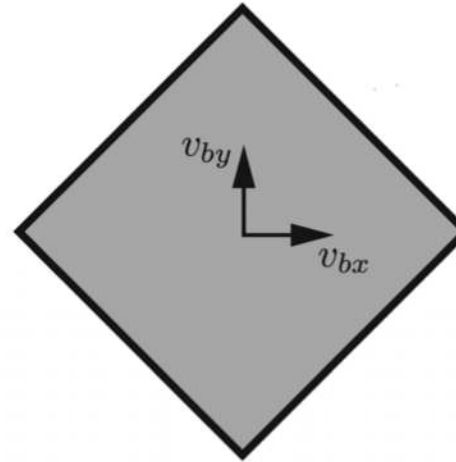
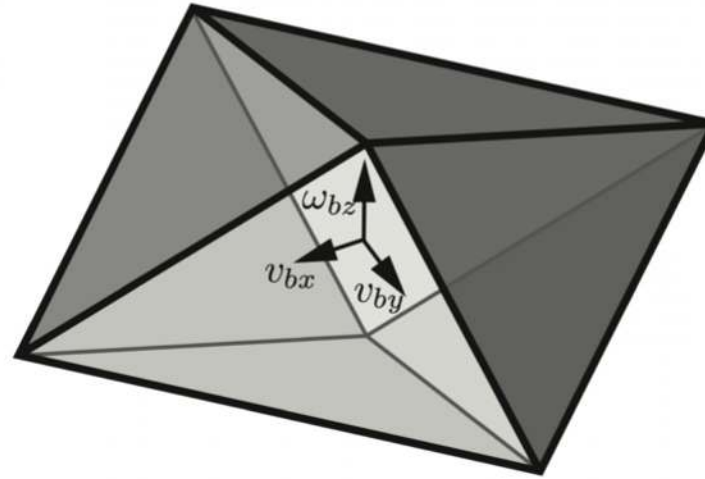
Important concepts, symbols, and equations (cont.)



Important concepts, symbols, and equations (cont.)

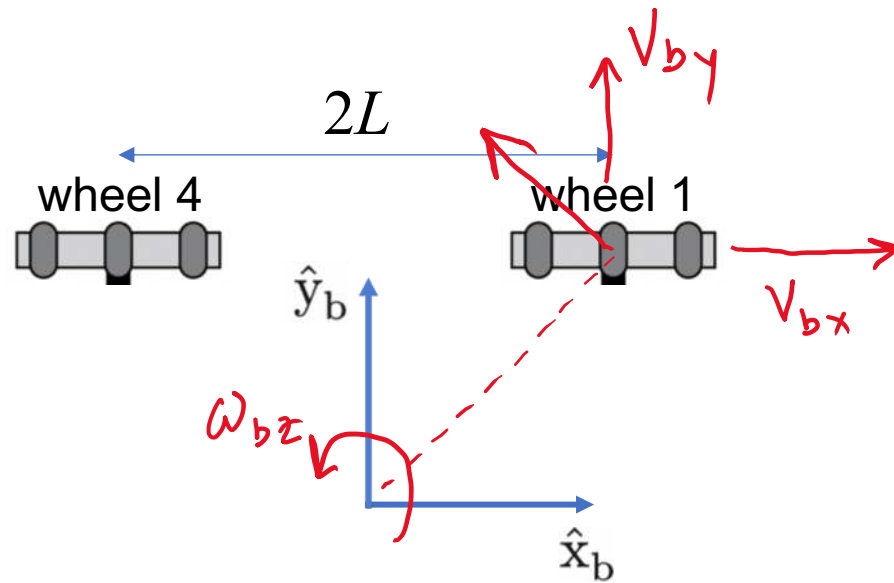


4 mecanum wheels

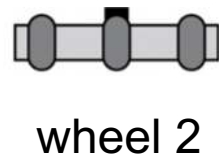
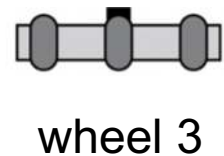


What does it mean
if the convex
polyhedron is
unbounded?

The $H(o)$ matrix has rank
2 or less. Some twists
cannot be obtained by
controlling wheel speeds.



The centers of the omniwheels of a mobile robot are at the corners of a square a distance $2L$ from each other. The radius of the wheels is r , and the forward driving direction for each wheel is to the right. What is $H(0)$?



$h_1(0)$ first. Find driving speed for $\begin{bmatrix} \omega_{bz} \\ v_{bx} \\ v_{by} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ or $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$h_1(0) = (-L, 1, 0)$

similarly

$$H(0) = \begin{bmatrix} h_1(0) \\ h_2(0) \\ h_3(0) \\ h_4(0) \end{bmatrix} = \frac{1}{r} \begin{bmatrix} -L & 1 & 0 \\ L & 1 & 0 \\ L & 1 & 0 \\ -L & 1 & 0 \end{bmatrix}$$

only rank 2
no way to achieve

$$\begin{bmatrix} 0 \\ 0 \\ v_{by} \end{bmatrix} = v_b$$